Numerical Methods

By

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Chapter (3): Interpolation and Extrapolation

In many scientific and engineering problems, the exact functional relationship between variables may not be known. Instead, the values of a function are often obtained from experimental measurements or observations and are presented in the form of a table.

However, in practical situations we may need to determine the value of the function at a point that is not included in the given data. To estimate such values, numerical techniques called interpolation and extrapolation are used.

Interpolation

- Interpolation is the process of estimating the value of a function within the range of the known data points.
- For example, if the values of a function are known at $x = 1,2,3,4$, and we want to estimate the value at $x = 2.5$, this is called **interpolation** because the value lies between two known data points.
- Interpolation is widely used in numerical analysis to construct a function that passes through the given data points and allows us to estimate intermediate values.

Extrapolation

- Extrapolation is the process of estimating the value of a function outside the range of the given data points.
- For example, if the data are given for $x = 1,2,3,4$, and we want to estimate the value at $x = 5$, this is an **extrapolation** problem.
- Extrapolation is generally less accurate than interpolation because it predicts values beyond the known data range.

Importance of Interpolation and Extrapolation

Interpolation and extrapolation are important tools in numerical methods and are widely used in fields such as:

- Engineering
- Physics
- Statistics
- Computer science
- Data analysis

They help in estimating unknown values when only limited data are available.

Importance of Interpolation and Extrapolation in Computer Science

These methods are used in many computer applications.

- **Computer Graphics:**

Interpolation is used to make images and animations look smooth.

Examples:

- Creating smooth animation between frames
- Coloring pixels in images
- Drawing curves and shapes

Image Processing:

Interpolation is used when resizing images.

Examples:

- Zooming in or zooming out of pictures
- Improving image quality

The computer estimates new pixel values using interpolation.

Data Analysis:

In data analysis, interpolation helps to estimate missing values in data.

Example:

- If some data values are missing, interpolation can approximate them.

Prediction:

Extrapolation is used to predict future values based on known data.

Examples:

- Predicting future sales
- Forecasting weather data

Contents of the Lecture:

1. Finite Difference Operators
2. Newton's forward interpolation formula for equal intervals

Finite Difference Operators

In numerical methods, finite difference operators are used to measure the changes between successive values of a function.
They are very important in interpolation, numerical differentiation, and numerical solutions of differential equations.
When the values of a function are given in a table form with equal intervals, finite differences help us analyze how the function changes from one point to another.

Forward Difference Operator

The forward difference or simply difference operator is denoted by Δ and may be defined as:

$$\Delta f(x) = f(x+h) - f(x)$$

or writing in terms of y,

$$\Delta y_i = y_{i+1} - y_i, i = 0,1,2,3,\dots,\mathrm{n}-1$$

The differences of the first differences are called **the second differences** and they are denoted by:

$$\Delta^2 y_0 = \Delta(\Delta y_0) = \Delta y_1 - \Delta y_0 = y_2 - y_1 - y_1 + y_0$$

$$\Delta^2 y_0 = y_2 - 2y_1 + y_0$$

$$\Delta^2 y_1 = \Delta(\Delta y_1) = \Delta y_2 - \Delta y_1 = y_3 - y_2 - y_2 + y_1$$

$$\Delta^2 y_1 = y_3 - 2y_2 + y_1$$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0					
		Δy_0				
x_1	y_1		$\Delta^2 y_0$			
		Δy_1		$\Delta^3 y_0$		
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$	
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$	
		Δy_3		$\Delta^3 y_2$		
x_4	y_4		$\Delta^2 y_3$			
		Δy_4				
x_5	y_5					

Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
$x_{-2} = x_0 - 2\mathrm{h}$	y_{-2}				
		Δy_{-2}			
$x_{-1} = x_0 - \mathrm{h}$	y_{-1}		$\Delta^2 y_{-2}$		
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$
		Δy_0		$\Delta^3 y_{-1}$	
$x_1 = x_0 + \mathrm{h}$	y_1		$\Delta^2 y_0$		
		Δy_1			
$x_2 = x_0 + 2\mathrm{h}$	y_2				

central difference table

Backward Difference Operator

The backward difference operator is denoted by ∇ and it is defined as

$$\nabla f(x) = f(x) - f(x-h)$$

or writing in terms of y,

$$\nabla y_i = y_i - y_{i-1}, \qquad i = \mathrm{n}, \mathrm{n}-1, \dots, 1$$

The differences of the first differences are called **the second differences** and they are denoted by:

$$\nabla^2 y_2 = \nabla(\nabla y_2) = \nabla y_2 - \nabla y_1 = y_2 - y_1 - y_1 + y_0$$

$$\nabla^2 y_0 = y_2 - 2y_1 + y_0$$

$$\nabla^2 y_3 = \nabla(\nabla y_3) = \nabla y_3 - \nabla y_2 = y_3 - y_2 - y_2 + y_1$$

$$\nabla^2 y_3 = y_3 - 2y_2 + y_1$$

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
x_0	y_0					
		∇y_1				
x_1	y_1		$\nabla^2 y_2$			
		∇y_2		$\nabla^3 y_3$		
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$	
		∇y_3		$\nabla^3 y_4$		$\nabla^5 y_5$
x_3	y_3		$\nabla^2 y_4$		$\nabla^4 y_5$	
		∇y_4		$\nabla^3 y_5$		
x_4	y_4		$\nabla^2 y_5$			
		∇y_5				
x_5	y_5					

Backward difference table (the horizontal table)

Example :

Construct the forward difference table and the horizontal table for the following data:

x	1	2	3	4	5
y	4	6	9	12	17

Solution : Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	4				
		2			
2	6		1		
		3		-1	
3	9		0		3
		3		2	
4	12		2		
		5			
5	17				

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	4				
		2			
2	6		1		
		3		-1	
3	9		0		3
		3		2	
4	12		2		
		5			
5	17				

Backward difference table (the horizontal table)

Example :

If $f\ (x)$ is known at the following data points.

x	45	55	65	75
$f(x)$	20	60	120	180

Construct a forward and backward difference tables

Solution :

Try by yourself

Newton’s forward interpolation formula

Newton’s Forward Interpolation Formula is used to estimate the value of a function when:

- The independent variable values are equally spaced.
- The required value of x is near the beginning of the table.

It uses forward differences to construct an interpolating polynomial.

Newton's forward interpolation formula

Equal Interval Condition:

The values of x must be equally spaced.

$$x_0,\ x_1,\ x_2,\ x_3, \ldots$$

Where

$$x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = \mathrm{h}$$

Here:

h = constant interval between x-values.

Newton's forward interpolation formula

Newton Forward Interpolation Formula:

$$f(x) = y_0 + \text{p}\,\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

Where $p = \frac{x - x_0}{h}$

and

x_0 is the first value in the table

h = equal spacing

Δy_0 = forward differences.

Example :

Find $y = e^{3x}$ for $x = 0.05$ using the following table.

x	0	0.1	0.2	0.3	0.4
y	1	1.3499	1.8221	2.4596	3.3201

Solution : Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	1				
		0.3499			
0.1	1.3499		0.1224		
		0.4723		0.0428	
0.2	1.8221		0.1652		0.0150
		0.6375		0.0578	
0.3	2.4596		0.2230		
		0.8605			
0.4	3.3201				

We have $x = 0.05, x_0 = 0, \mathrm{h} = 0.1$

$$p = \frac{x - x_0}{h} = \frac{0.05 - 0}{0.1} = 0.5$$

Using Newton's forward formula

$$f(x) = y_0 + \mathrm{p}\,\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$f(0.05)$$

$$= 1 + (0.5)(0.3499) + \frac{(0.5)(0.5-1)}{2!}(0.1224)$$

$$+ \frac{(0.5)(0.5-1)(0.5-2)}{3!}(0.0428)$$

$$+ \frac{(0.5)(0.5-1)(0.5-2)(0.5-3)}{4!}(0.0150) = 1.16172$$

Example :

If $f\ (x)$ is known at the following data points.

x	0	1	2	3	4
$f(x)$	1	7	23	55	109

Find $f(0.5)$ using Newton's forward difference formula.

Solution : Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	1				
		6			
1	7		10		
		16		6	
2	23		16		0
		32		6	
3	55		22		
		54			
4	109				

We have $x = 0.5, x_0 = 0, \mathrm{h} = 1$

$$p = \frac{x - x_0}{h} = \frac{0.5 - 0}{1} = 0.5$$

Using Newton's forward formula

$$f(x) = y_0 + \mathrm{p}\,\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$f(0.5)$$
$$= 1 + (0.5)(6) + \frac{(0.5)(0.5-1)}{2!}(10) + \frac{(0.5)(0.5-1)(0.5-2)}{3!}(6)$$
$$+ \frac{(0.5)(0.5-1)(0.5-2)(0.5-3)}{4!}(0) = 3.125$$

Example :

The profits of a company (in thousands of rupees) are given below:

Year x	1990	1993	1996	1999	2002
Profit $f(x)$	120	100	111	108	99

Calculate the total profits between 1990–2002. calculate profits at 1991.

Solution : Forward difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1990	120				
		-20			
1993	100		31		
		11		-45	
1996	111		-14		53
		-3		8	
1999	108		-6		
		-9			
2002	99				

We have $x = 1991, x_0 = 1990, \mathrm{h} = 3$

$$p = \frac{x - x_0}{h} = \frac{1991 - 1990}{3} = 0.33$$

Using Newton's forward formula

$$f(x) = y_0 + \mathrm{p}\,\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$f(0.5)$$
$$= 120 + (0.33)(-20) + \frac{(0.33)(0.33-1)}{2!}(31)$$
$$+ \frac{(0.33)(0.33-1)(0.33-2)}{3!}(-45)$$
$$+ \frac{(0.33)(0.33-1)(0.33-2)(0.33-3)}{4!}(53) = 104.93$$

Example :

If $f\ (x)$ is known at the following data points.

x	40	45	50	55	60
$f(x)$	0.6428	0.7071	0.7660	0.8192	0.8660

Find $f(42)$ using Newton's forward difference formula.

Solution :

Try by yourself


THANK YOU
FOR YOUR
ATTENTION